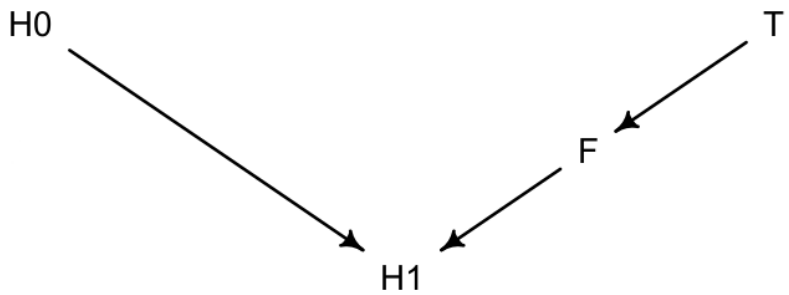
Overfitting and towards conditioning

So far

- Model comparison (don't "select") is not causal inference.
- Multiple models can inform causal inference.
- Multiple models can be competing to explain, in the prediction sense.

Model mis-selection (revisited)

- Recall the plant/fungus example from Chapter 6 in the SR book.
- Three models:
  - `m6.6`: intercept only
  - `m6.7`: Treatment+Fungus
  - `m6.8`: Treatment only




```
compare(m6.6, m6.7, m6.8)
compare(m6.6, m6.7, m6.8, func = PSIS)
```

Model mis-selection (revisited): Model comparison

```
compare(m6.6, m6.7, m6.8)
```

```
##           WAIC       SE    dWAIC      dSE    pWAIC        weight
## m6.7 361.8105 14.17580  0.00000       NA 3.785343 1.000000e+00
## m6.8 403.0649 11.23902 41.25445 10.42699 2.791505 1.100805e-09
## m6.6 406.2842 11.76437 44.47372 12.20547 1.752989 2.201174e-10
```

- WAIC: estimated out-of-sample log-score
- SE: standard error or WAIC
- dWAIC: difference from top model
- dSE: standard error of dWAIC
- pWAIC: penalty or "effective number of parameters
- weight: Akaike weight

Model mis-selection (revisited): Model comparison

```
compare(m6.6, m6.7, m6.8, func = PSIS)
```

```
##           PSIS       SE    dPSIS      dSE    pPSIS        weight
## m6.7 361.5462 14.12214  0.00000       NA 3.610101 1.000000e+00
## m6.8 403.1572 11.38878 41.61095 10.27510 2.826919 9.210784e-10
## m6.6 406.0888 11.79778 44.54263 12.06853 1.669705 2.126621e-10
```

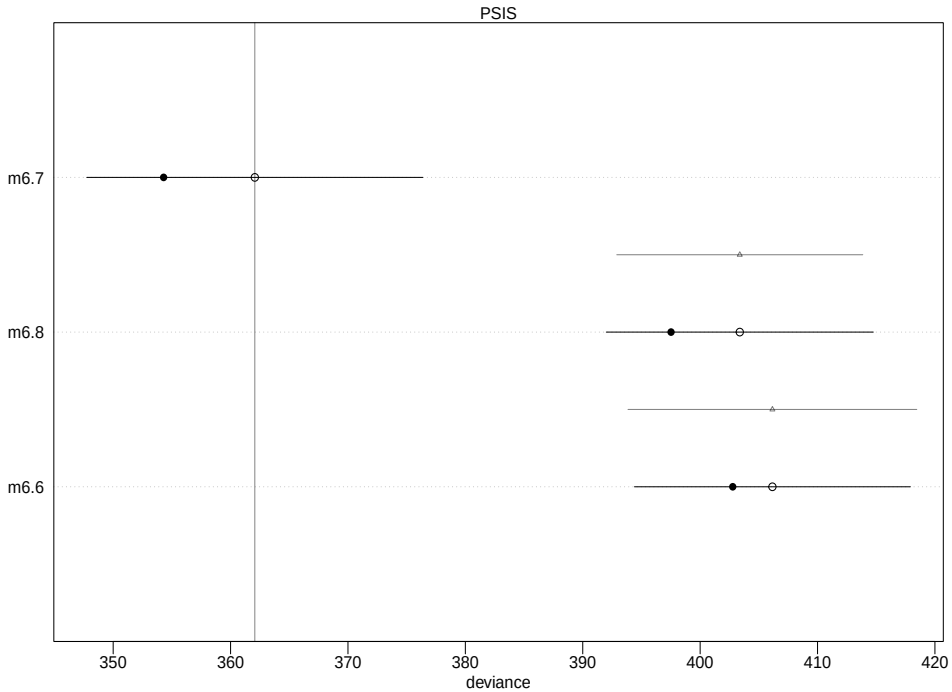
- PSIS: estimated out-of-sample log-score
- SE: standard error or PSIS
- dPSIS: difference from top model
- dSE: standard error of dPSIS
- pPSIS: penalty or "effective number of parameters
- weight: Akaike weight

Model mis-selection (revisited): Model comparison

```
compare(m6.6, m6.7, m6.8, func = PSIS)
```

```
##          PSIS       SE    dPSIS      dSE    pPSIS       weight
## m6.7 361.8603 14.19627  0.00000       NA 3.744065 1.000000e+00
## m6.8 402.8723 11.26119 41.01208 10.35612 2.719644 1.242626e-09
## m6.6 406.3034 11.79815 44.44313 12.23217 1.763768 2.235101e-10
```

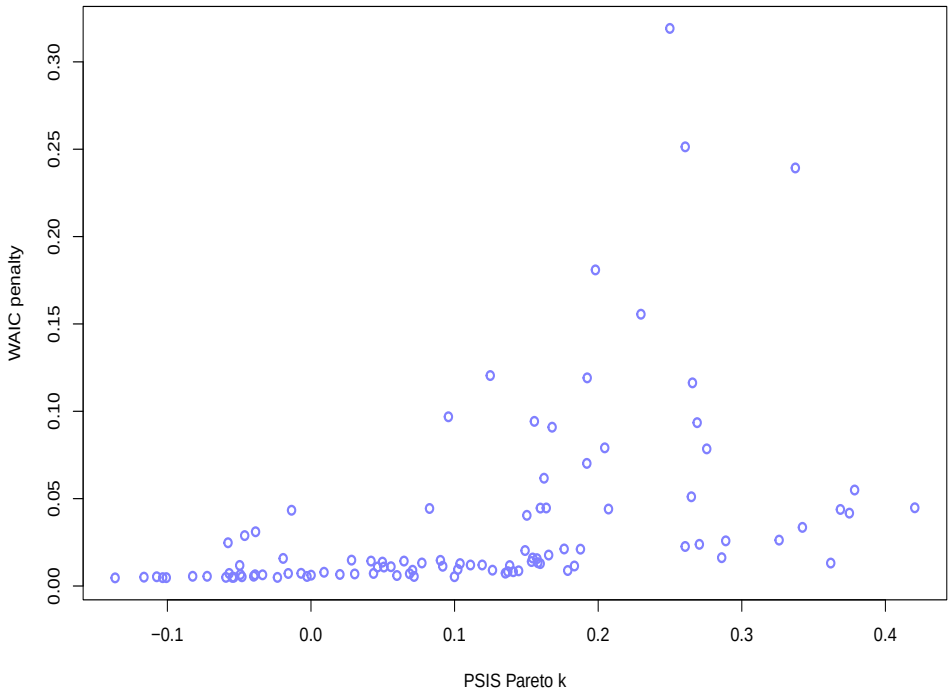
```
plot( compare(m6.6, m6.7, m6.8, func = PSIS) )
```



Model mis-selection (revisited): Model comparison

```
PSIS_m6.7 <- PSIS(m6.7, pointwise = T)
WAIC_m6.7 <- WAIC(m6.7, pointwise = T)

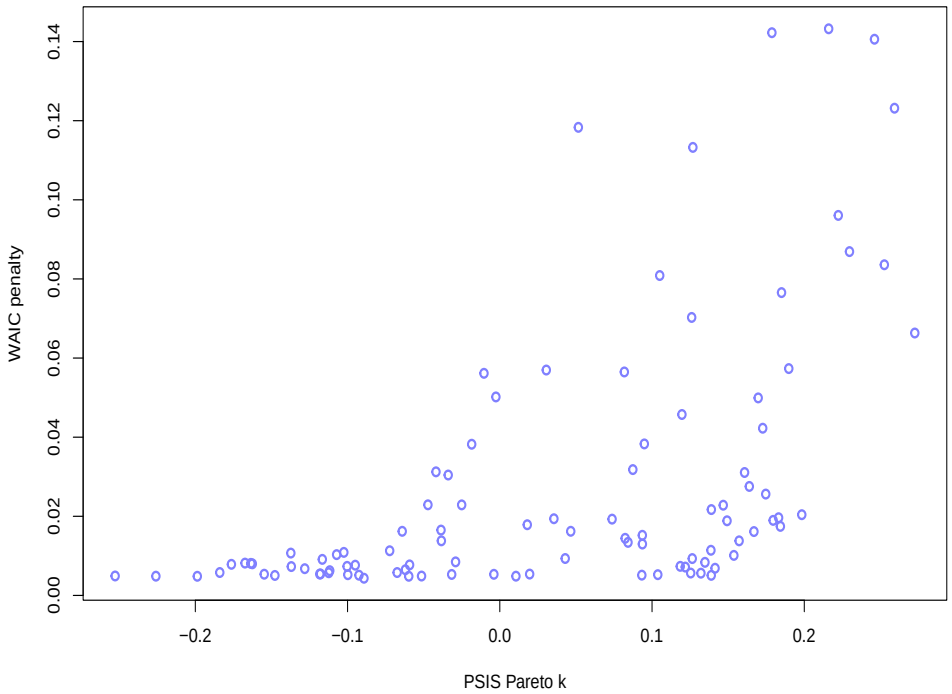
plot( PSIS_m6.7$k, WAIC_m6.7$penalty, xlab = "PSIS Pareto k",
    ylab = "WAIC penalty", col = rangi2, lwd = 2 )
```



Model mis-selection (revisited): Model comparison

```
PSIS_m6.8 <- PSIS(m6.8, pointwise = T)
WAIC_m6.8 <- WAIC(m6.8, pointwise = T)

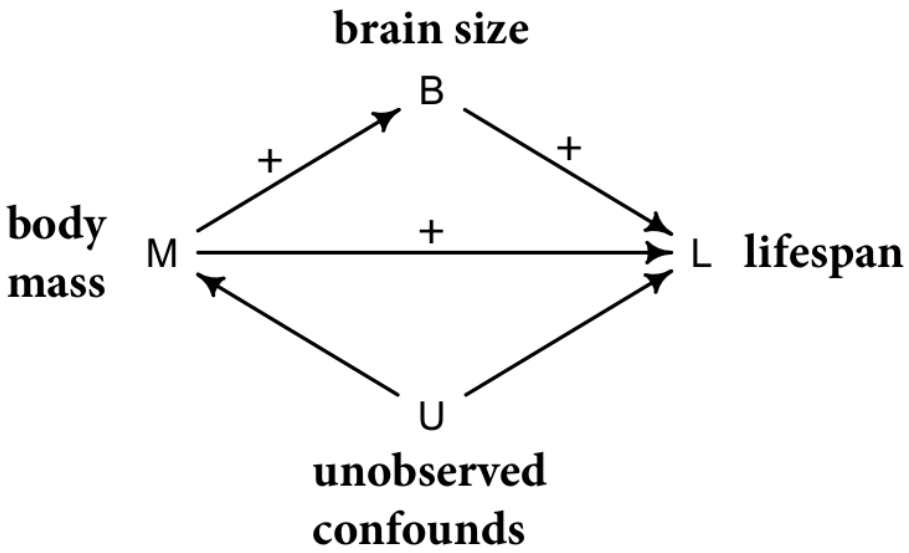
plot( PSIS_m6.8$k, WAIC_m6.8$penalty, xlab = "PSIS Pareto k",
    ylab = "WAIC penalty", col = rangi2, lwd = 2 )
```

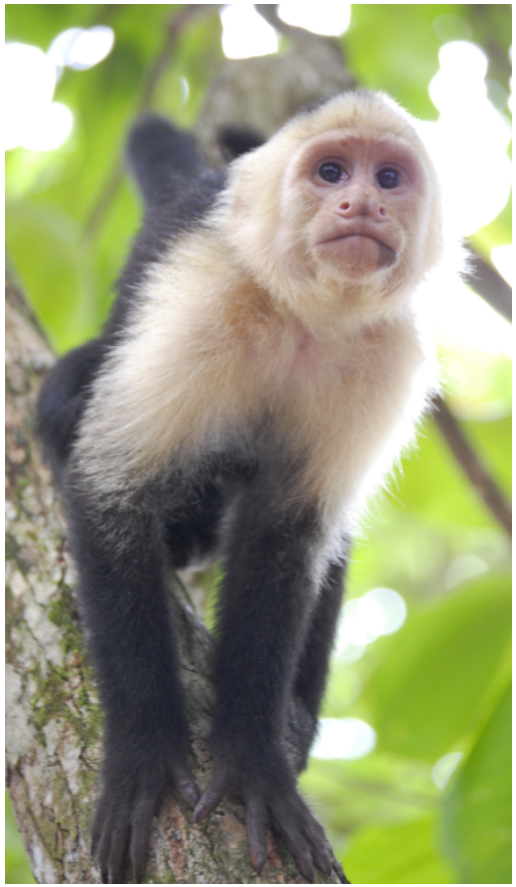


Note: it is not worth going for robust regression here

Something about monkeys

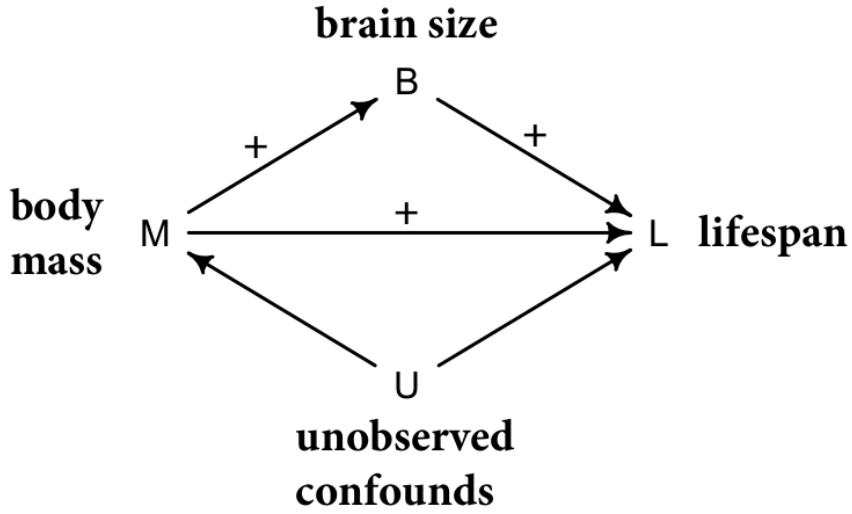
- Why do primates live a long time?
- Consider:

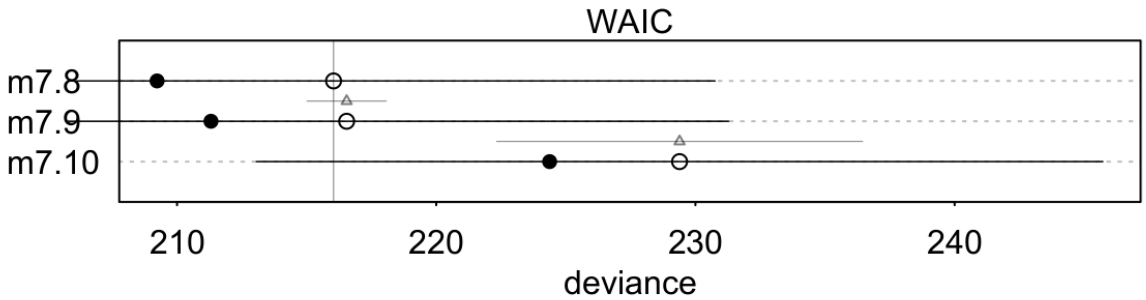




Something about monkeys

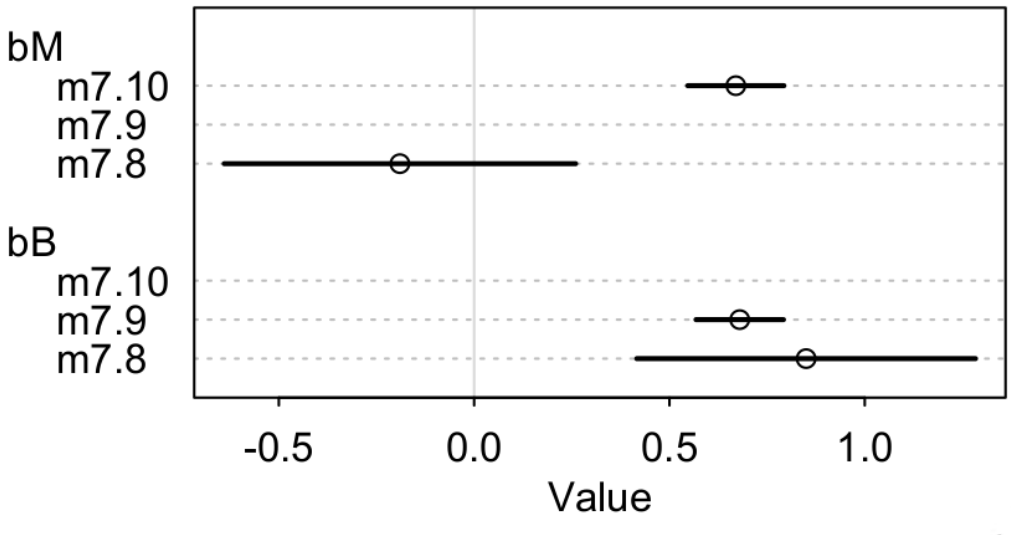
- 112 primate species
- Three models
  - m7.8: $\log L \sim \log M + \log B$
  - m7.9: $\log L \sim \log B$
  - m7.10: $\log L \sim \log M$
- What happens?





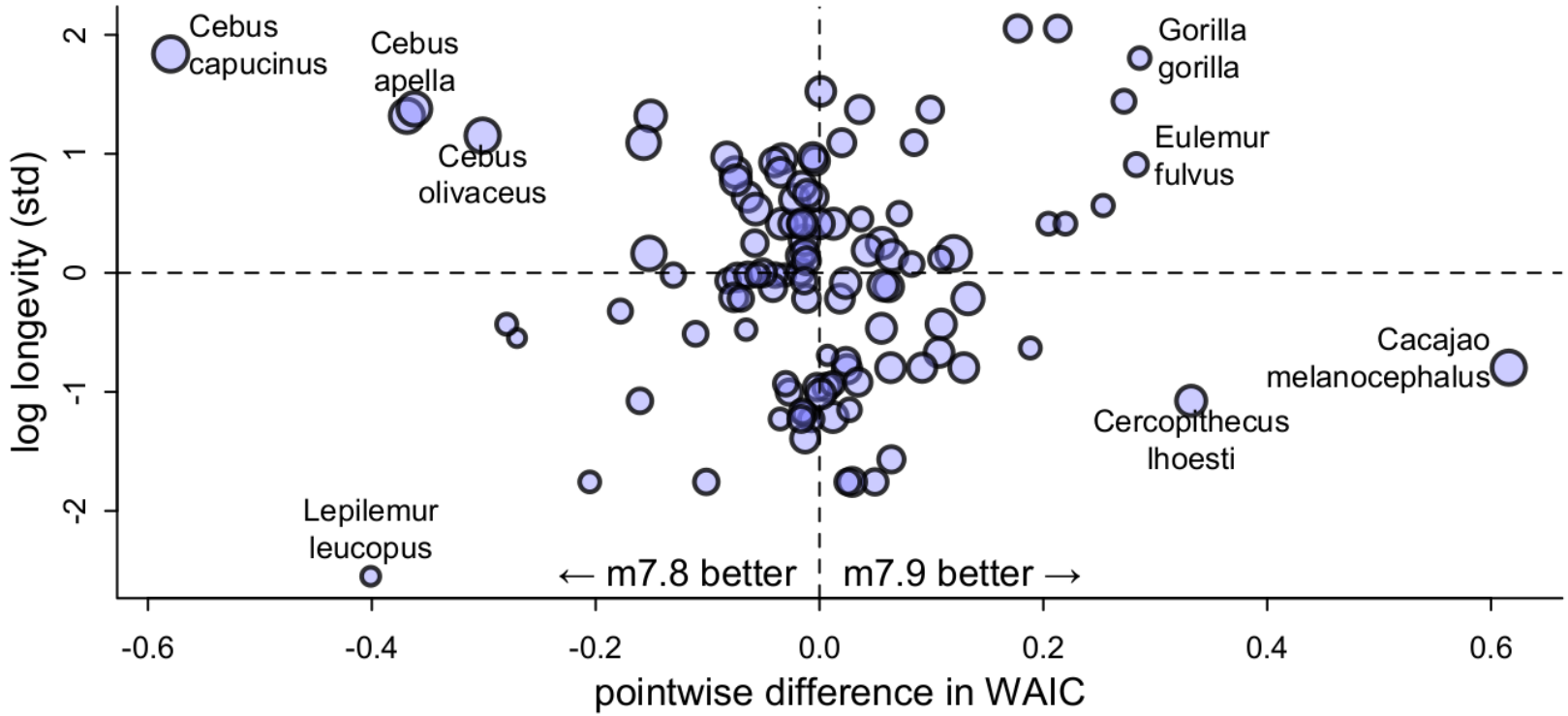
Something about monkeys

- Inspect parameter posterior distributions
- Why does body mass' effect go negative in the joint model?



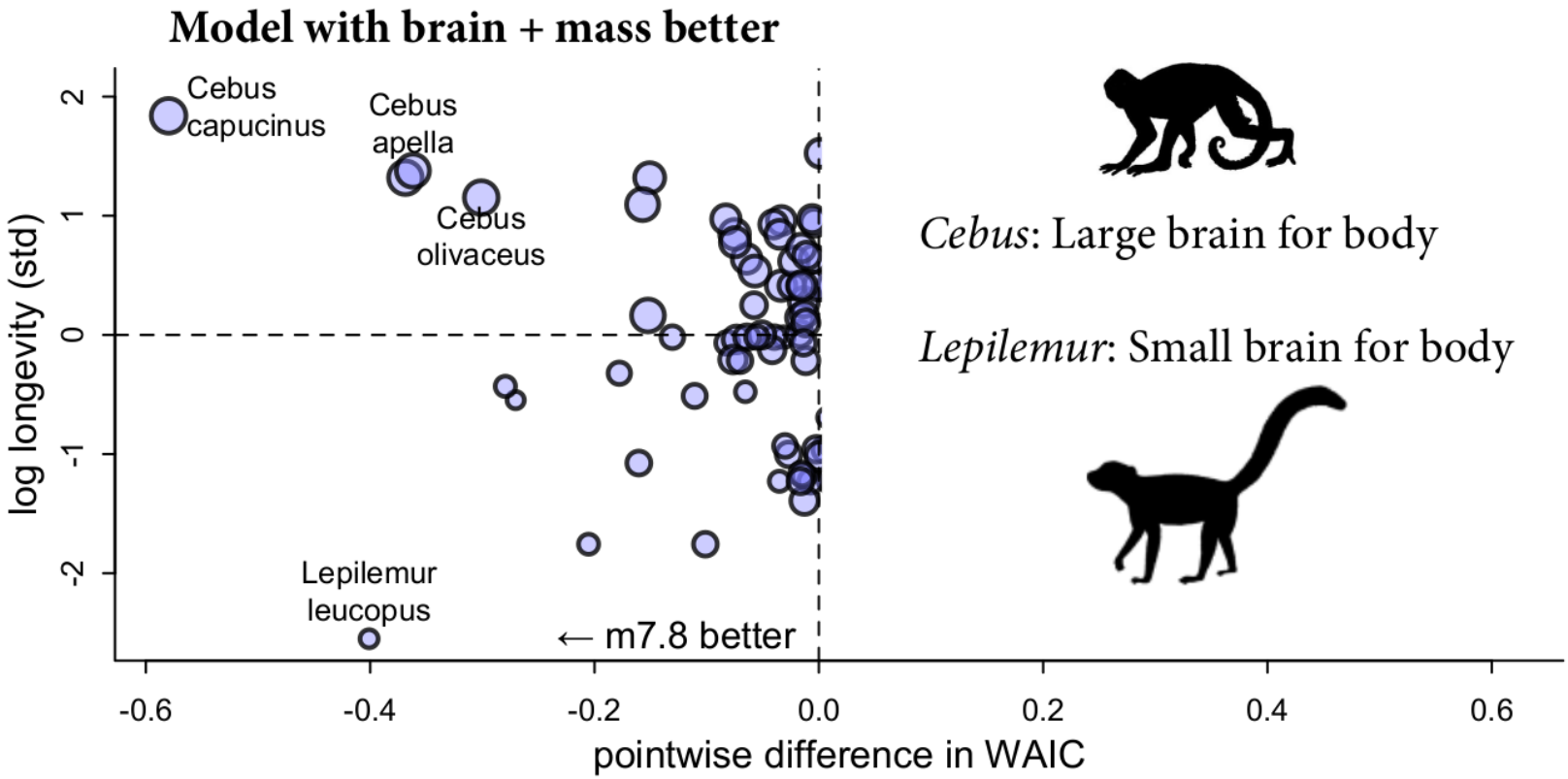


Something about monkeys: pointwise perspective



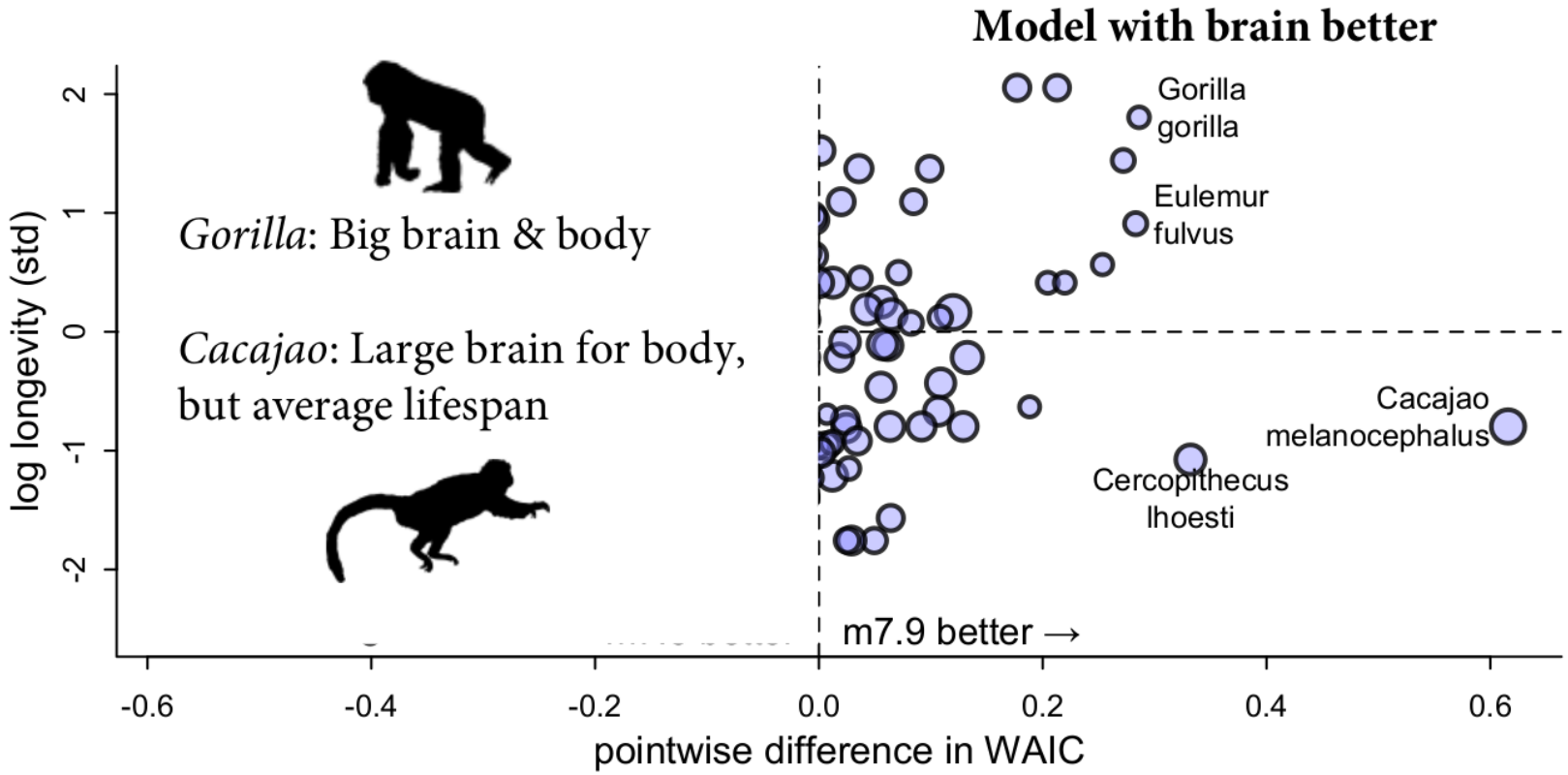
Point size proportional to abs diff brain z-score – body z-score

Something about monkeys: pointwise perspective



Point size proportional to abs diff brain z-score – body z-score

Something about monkeys: pointwise perspective

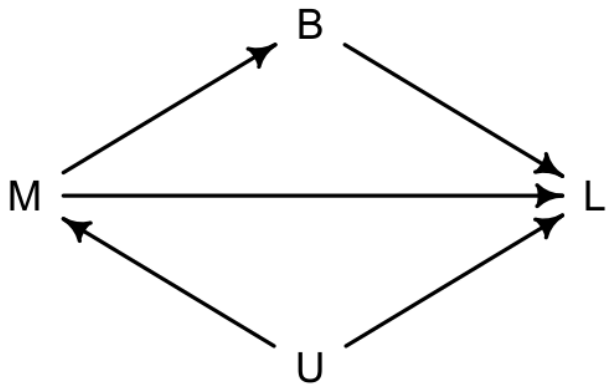


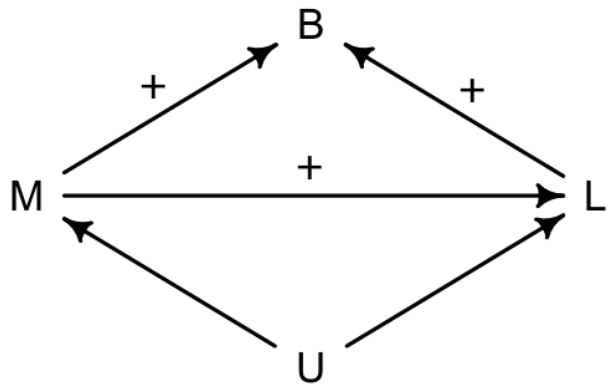
Point size proportional to abs diff brain z-score – body z-score

Monkey collider



- Another idea: reversal of body size coefficient consistent with collider bias
- Conditioning on brain size would open the $M \rightarrow B \leftarrow L$ backdoor path.

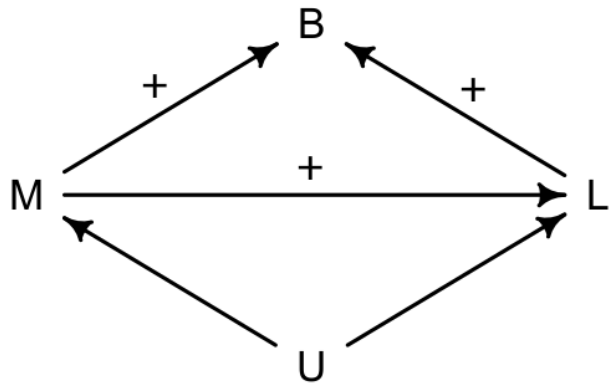




Monkey collider

```
m7.11 <- quap(alist(
  log_B ~ dnorm(mu, sigma),
  mu <- a+bM*logM+bL*log_L,
  a ~ dnorm(0, 0.1),
  c(bM, bL) ~ dnorm(0, 0.5),
  sigma ~ dexp(1)
), data = monkeys)

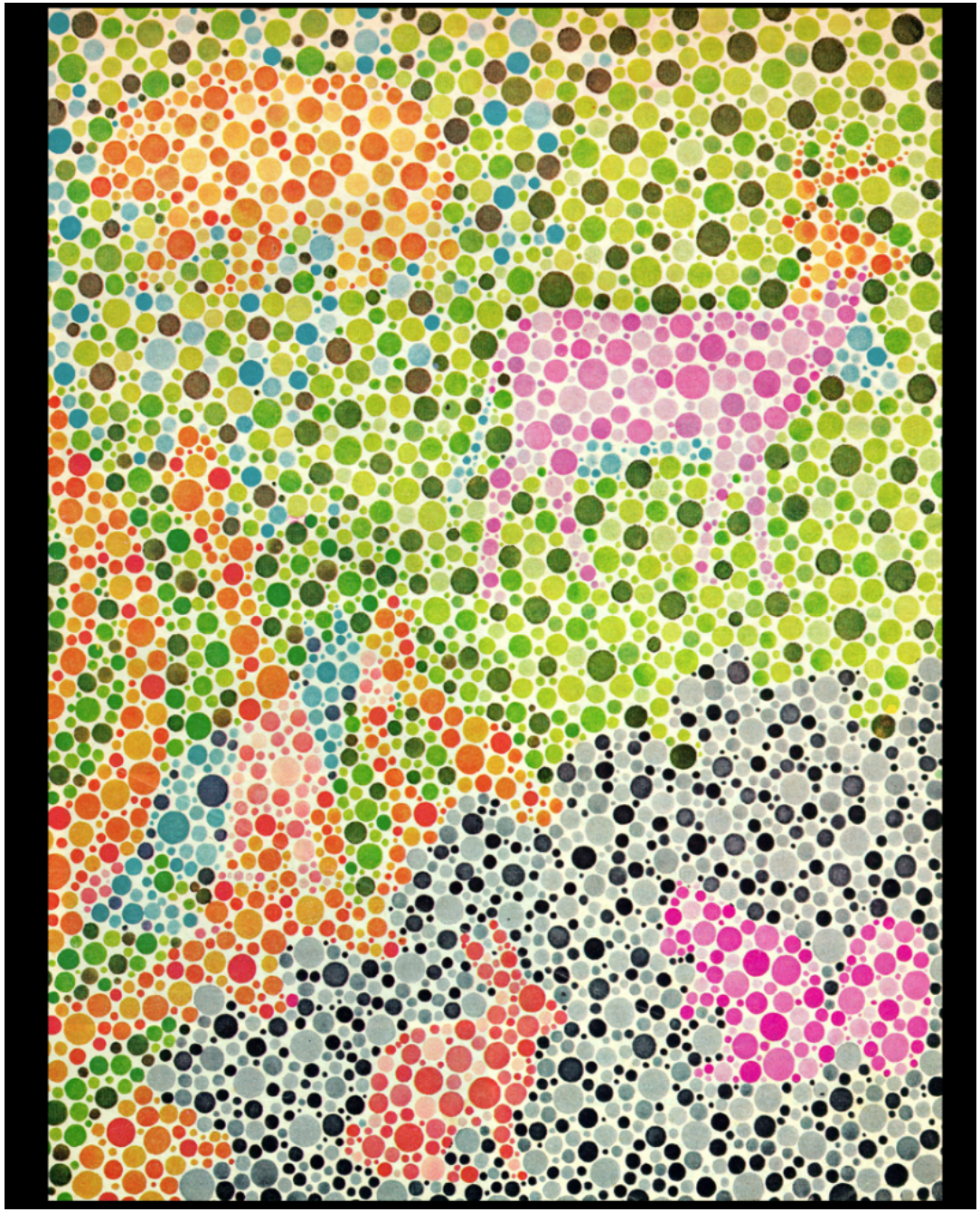
precis(m7.11)
```




```
mean   sd  5.5%  94.5%
```

a -0.05 0.02 -0.07 -0.02
bM 0.94 0.03 0.90 0.98
bL 0.12 0.03 0.07 0.16
sigma 0.19 0.01 0.17 0.21

Only small examples of what is possible. Trying all possible models is a formula for overfitting. Be thoughtful and honest in admitting data exploration.



Stop testing, start thinking

- Avoid off-the-shelf tools
  - Already made tests and p-values
  - Information criteria for model selection
  - Linear models only
- Good decision benefits from bespoke tools, e.g. in risk analysis or model design.

bespoke | bə'spōk |

adjective *chiefly British*

made for a particular customer or user: *a bespoke suit* | *bespoke kitchens* | *bespoke software systems* | *group tours and bespoke itineraries*.

- making or selling bespoke goods, especially clothing: *bespoke tailors*.

N.Y. / REGION

1350 COMMENTS

Leaders in New York and New Jersey Defend Shutdown for a Blizzard That Wasn't

By MATT FLEGENHEIMER JAN. 27, 2015

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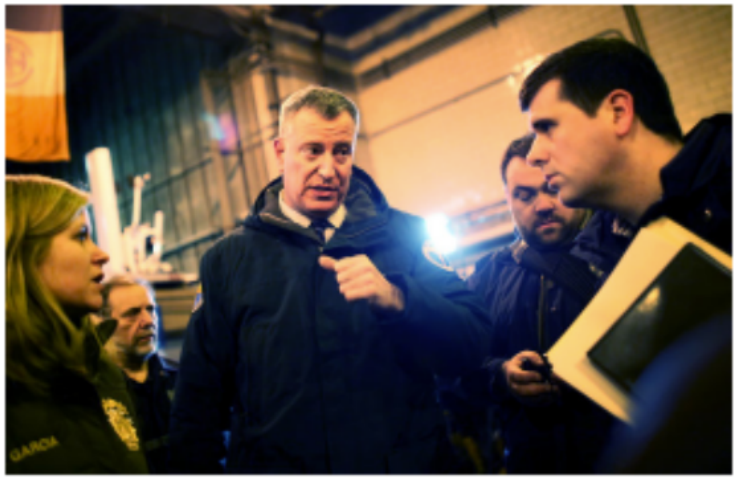
Save

More

It was an unprecedented step for what became, in New York City, a common storm: For the first time in its 110-year history, the subway system was shut down because of snow.

Transit workers, caught off guard by the shutdown that Gov. Andrew M. Cuomo announced on Monday, scrambled to grind the network to a halt within hours.

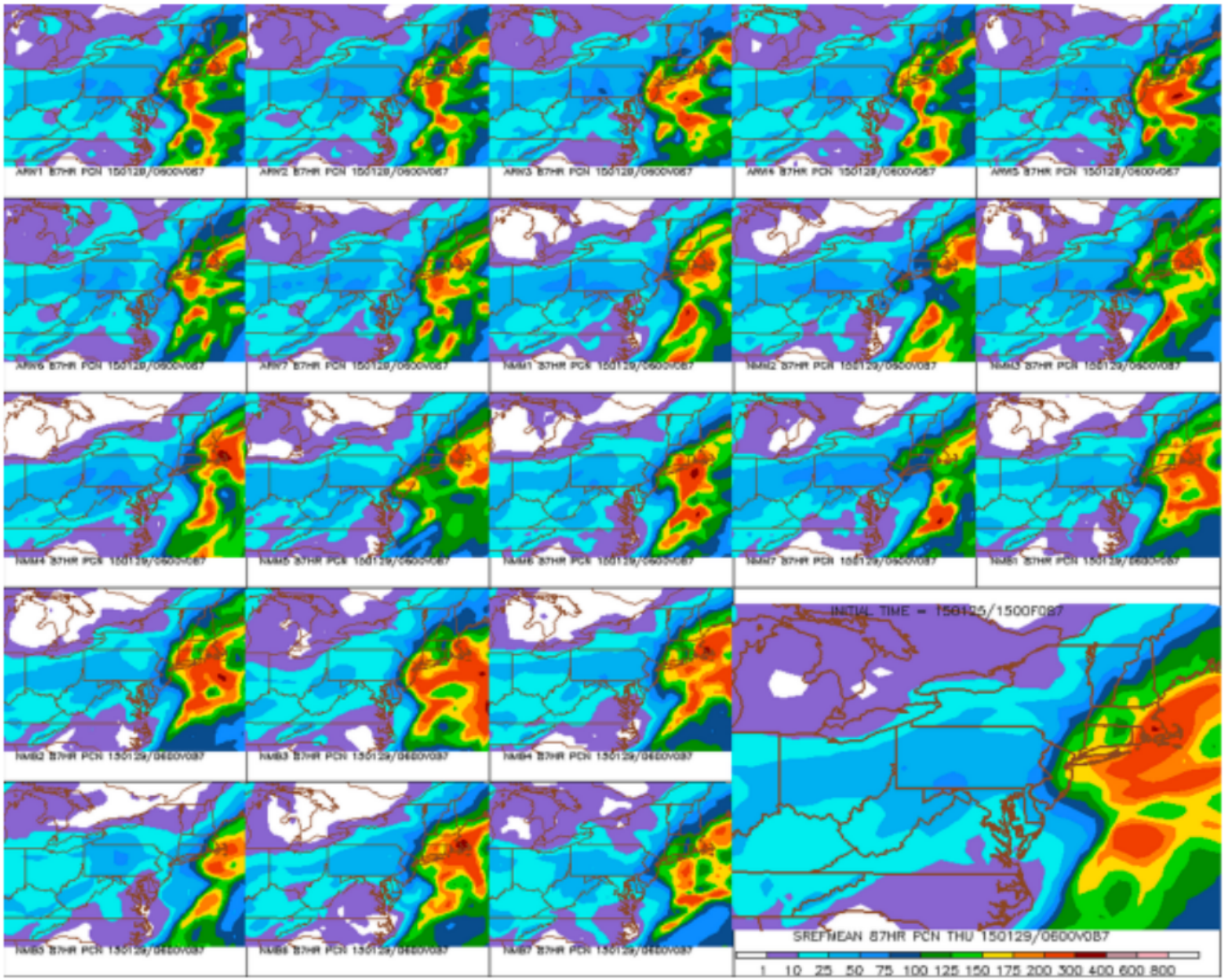
Residents moved quickly to find places



Mayor Bill de Blasio of New York City, with Sanitation Department workers in Manhattan on Monday, when he issued dire warnings about the storm. Yana Paskova for The New York Times

ECMWF About Forecasts

World leader in global medium-range numerical weather prediction



Blizzard calibration

- Was it bad to predict NYC blizzard from ECMWF model?
- Other models were more accurate.
- But welfare enhanced by being prepared $\rightarrow$ use extreme forecasts
- Accuracy always matters. BUT it isn't all that matters.



Conditional manatees



Conditional plane damage



How is manatee like an A.W.38 bomber?

